Meal Plan Generation in Vietnamese Cuisine using Dynamic Programming and Scientifically-Grounded Nutritional Profiling

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Abstract. The paradigm of personalized nutrition demands intelligent systems that can translate individual health data into actionable dietary guidance. However, existing Food Recommendation Systems (FRS) are often limited by a lack of cultural specificity and scientific rigor, typically failing to generate complete, nutritionally-balanced daily meal plans. This paper addresses these gaps by introducing a novel computational framework for automated meal plan generation, specifically tailored to Vietnamese cuisine. We present three key contributions: First, we detail a replicable pipeline using Large Language Models (LLMs) to construct a robust, culturally-specific knowledge base from unstructured Vietnamese cookbooks and heterogeneous nutritional sources. Second, we implement a scientifically-grounded user profiling module that creates precise nutritional targets by systematically applying established standards, including the Mifflin-St Jeor equation for Basal Metabolic Rate (BMR), Total Daily Energy Expenditure (TDEE), and official Recommended Dietary Allowances (RDA) for comprehensive micronutrient coverage. Finally, to overcome the "planning gap" inherent in single-recipe recommenders, we propose a two-phase optimization algorithm that leverages Dynamic Programming to efficiently synthesize holistic 3-meal daily plans that satisfy a user's complex macro- and micronutrient requirements. This work represents a significant step toward creating more practical, personalized, and effective dietary support tools.
Keywords: Personalized Nutrition, Food Recommendation Systems, Dynamic Programming

1 Introduction

The global discourse on health and wellness has undergone a fundamental transformation in recent decades. A confluence of factors, including heightened consumer health consciousness, the escalating prevalence of diet-related chronic diseases, and the pervasive integration of digital technologies into daily life, has catalyzed a significant paradigm shift in nutritional science [1]. This shift marks a departure from generalized, population-level dietary advice towards a more nuanced, individualized approach known as personalized nutrition [2].

1.1 The Inevitability of Adopting Personalized Nutrition

In the preceding decades, public health nutrition was dominated by generic guidelines, exemplified by food pyramids and standardized daily allowances. These models, while beneficial for establishing broad public health goals, often fail to produce consistent, long-term results at the individual level [3]. Specifically, generic diets are typically rigid, prescribing strict rules that may not align with an individual's unique tastes, lifestyle, or daily schedule. Hence, generic nutrition is difficult to sustain over time and frequently leads to burnout [3]. More fundamentally, the vast heterogeneity in human biology, including metabolic rates, genetic predispositions, and physiological responses to specific nutrients, remains absent from those generalized nutrition guidelines. Furthermore, generic plans inadequately account for the individual's health conditions such as diabetes and food allergies.

In response to these limitations, personalized nutrition has emerged as a scientifically robust alternative. This approach tailors dietary recommendations to an individual's specific profile, which can encompass a wide range of data points including biometrics (age, weight, height), physiological state (health conditions, activity levels), inherent gene and metabolism [2]. By aligning dietary advice with these unique characteristics, personalized nutrition holds the promise of dramatically improved health outcomes, enhanced motivation, and a powerful strategy for the prevention of chronic, diet-related diseases [1]. Moreover, the proliferation of wearable technology, health-tracking applications, and direct-to-consumer genetic testing has made the collection of personalized data more feasible. Therefore, this data-driven mandate facilitates the requirements for systems that can not only collect and aggregate this multifaceted personal data but also interpret it and translate it into practical, actionable, and daily dietary guidance.

1.2 Food Recommendation Systems for Personalized Dietary Guidance

Food Recommendation Systems (FRS) have emerged as the key enabling technology to meet this data-driven mandate. Formally, an FRS is a specialized information filtering system designed to produce individualized recommendations, guiding a user toward relevant, or useful items within the vast and often overwhelming space of food choices [4]. With the explosion of online food recipes from blogs, social media, and cookbooks, FRS has become an essential tool for helping users navigate information overload and discover new meals [5].

While within the health and wellness field, FRS has demonstrated its significant impacts on promoting individual lifestyles [4]. The application of recommender systems to the nutritional domain introduces a layer of complexity. Specifically, a health-aware FRS has to resolve challenging multi-objective optimization problems of simultaneously balancing the user's subjective preference (such as savoriness, cuisine types, and convenience (e.g., preparation time)) with user's health-based constraints (such as caloric targets, specific macronutrient ratios, and micronutrient requirements) [6].

Notably, the system described in this paper explicitly models this duality, collecting both preferential data (e.g., disliked ingredients, maximum preparation time) and physiological data (e.g., age, weight, activity level) to generate a comprehensive user profile. Moreover, a successful health-aware FRS cannot simply recommend a user's

preferences, as it must simultaneously determine a user's needs. This inherent tension between preference and necessity is a central challenge that motivates the subsequently presented work.

1.3 The Cultural Contextualization Challenge

While the potential of FRS is significant, the vast majority of existing research and commercial systems have been developed and validated within the context of Western cuisines. This narrow focus leaves a substantial gap in the literature and an underserved global population. Food is a deeply cultural preference, as concepts of what is palatable or traditional vary dramatically across different societies [17]. Consequently, the currently presented FRS that recommends dishes without cultural context is likely to fail, as demonstrated by systems that suggest Thai or Chinese dishes to users seeking culturally specific Indian recipes [16].

1.4 Problem Statement, Proposed Framework, and Contributions

Given the established limitations of generic dietary guidelines, multi-objective optimization problems of FRS, and the significant gap in current FRS research concerning non-Western cuisines, specifically Vietnamese cuisines, there is a pressing need for a system capable of delivering nutritionally-precise, culturally-aware, and practically actionable meal recommendations. Existing systems often fall short in one or more of these areas: they may lack cultural specificity, provide recommendations that are difficult for a user to act upon, or rely on overly simplistic nutritional models. This is particularly true for Vietnamese cuisine, where the complexity and relative scarcity of structured data present significant barriers to entry.

Motivated by the above discourses, the core research problem addressed in this paper is therefore to design, implement, and evaluate a novel FRS framework that overcomes these challenges. This is achieved by first constructing a robust, computable knowledge base of Vietnamese cuisine and then developing a recommendation engine that evolves beyond passive matching to provide active, supplementary dietary guidance. To this end, we introduce a comprehensive computational framework for personalized nutritional recommendation tailored specifically to Vietnamese scenarios. The primary contributions of this paper are threefold:

1. **A Robust and Cultural Knowledge Base.** We detail a replicable data engineering pipeline that transforms unstructured Vietnamese cookbooks and heterogeneous nutritional databases (USDA and Vietnamese FCT) into a unified, analysis-ready knowledge base. This process leverages Large Language Models (LLMs) for automated data extraction and harmonization, providing a foundational resource for applying advanced AI to an under-resourced culinary domain.
2. **Scientifically-Grounded Nutritional Profiling.** We introduce a rigorous methodology for user-centric target modeling that translates qualitative user goals into a quantitative, multi-dimensional nutritional vector. This is achieved by systematically applying medically-established standards: the **Mifflin-St Jeor equation** is used to calculate Basal Metabolic Rate (BMR) and Total Daily Energy

Expenditure (TDEE) for precise caloric and macronutrient targets, while **Recommended Dietary Allowances (RDA)** are integrated to ensure comprehensive micronutrient coverage. This contribution ensures that recommendations are not arbitrary but are anchored in scientific validity.

3. **A Novel Two-Phase Meal Plan Optimization Framework.** As our principal technical contribution, we propose an advanced algorithm to solve the "planning gap." To manage the immense computational complexity of selecting a 3-meal plan that satisfies dozens of nutritional targets, our framework employs a two-phase strategy:
   - **Phase 1: Macro-Focused Dynamic Programming (DP).** The system first uses a DP algorithm to efficiently generate a set of candidate meal plans that are highly optimized for the user's core caloric and macronutrient goals.
   - **Phase 2: Holistic Scoring and Final Selection.** These candidates are then evaluated against the user's full nutritional profile, with a scoring function that penalizes any micronutrient deficiencies. This ensures the final recommended plan is not just calorically appropriate but holistically nutritious, representing a significant step towards truly practical and effective dietary support.

2 Related Works

To situate our contributions within the existing body of research, this section provides a comprehensive review of the literature on Food Recommendation Systems. We begin by examining the foundational algorithmic paradigms that underpin the field. We then trace the evolution of these systems toward health-aware applications and discuss the critical, yet often overlooked, challenge of knowledge base construction for culturally-specific domains. Finally, we synthesize these threads to identify a crucial "actionability gap" in the current state-of-the-art, which our proposed framework is uniquely designed to address.

2.1. Foundational Paradigms in Food Recommendation

The design of recommendation systems is primarily driven by three canonical paradigms: Content-Based (CB) filtering, Collaborative Filtering (CF), and Hybrid approaches that combine the two. These paradigms have been extensively applied and adapted within the food domain.

- **Content-Based (CB) Filtering** operates on the principle of recommending items that are similar to those a user has previously liked or shown interest in. This approach relies on the intrinsic properties or "content" of the items themselves. In the food domain, these properties can include ingredients, cuisine type, preparation method, flavor profiles, and, most importantly for health-aware systems, detailed nutritional information [9]. A CB system typically creates a profile for each user based on their preferences for these attributes and a profile for

each food item. Recommendations are then generated by matching item profiles to the user's profile using techniques such as vector space models (e.g., Term Frequency-Inverse Document Frequency, TF-IDF) and similarity metrics like cosine similarity [11]. The initial versions of our proposed methodology are fundamentally content-based, representing both the user's nutritional targets and each recipe's nutritional composition as vectors and using distance or similarity metrics to measure their alignment [15].

- **Collaborative Filtering (CF)** takes a different approach, leveraging the "wisdom of the crowd." Instead of analyzing item content, CF recommends items based on the preferences and behaviors of other, "similar" users [10]. The core idea is that if user A and user B have similar tastes (e.g., they have rated many of the same items similarly), then an item that user B liked but user A has not yet seen is a good candidate for recommendation. CF methods are typically divided into user-based and item-based approaches and rely on a large user-item interaction matrix, such as explicit ratings [11]. While powerful, CF systems famously suffer from the "cold-start" problem, where they cannot make effective recommendations for new users or new items with insufficient interaction data.
- **Hybrid Systems** seek to amalgamate CB and CF techniques to capitalize on their respective strengths while mitigating their weaknesses.[9] For example, a hybrid system might use a CB approach to handle new items (which have content but no ratings) and a CF approach for established items, or it might combine the scores from both recommenders using a weighted, switching, or cascading model [9]. This approach often leads to more robust and accurate recommendations by avoiding the over-specialization of CB systems and the cold-start problem of CF systems [29].

While all three paradigms are prevalent in the broader FRS landscape, health-aware systems exhibit a natural and logical inclination towards content-based logic. The core task in personalized nutrition is to match a food's objective properties (its nutritional "content") to a user's objective health requirements (caloric and nutrient targets). This is fundamentally a content-matching problem. Collaborative filtering, which relies on subjective taste similarity, is less suited for this primary task. For instance, two users may share identical health goals (e.g., a low-sodium, high-protein diet for managing hypertension) but possess entirely different culinary preferences. A CF system might erroneously recommend dishes based on taste overlap, failing to satisfy the preferences of one user. Therefore, the decision to build our framework upon a content-based foundation is a deliberate choice, aligned with the specific requirements of the health-aware recommendation sub-domain.

2.2. The Evolution Towards Health-Aware Systems

Building upon these foundational paradigms, a significant and growing body of research has focused on developing FRS specifically for health and wellness applications. These "health-aware" systems represent an evolution from simple preference-matching to more complex, constraint-based optimization.

A common approach in this domain involves the creation of a detailed user profile that goes beyond simple taste preferences to include physiological and health-related data. Many systems, including our own, begin by calculating a user's Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE) using validated formulas like the Mifflin-St Jeor equation, and then adjust these values based on the user's activity level and specific health goals (e.g., weight loss, muscle gain) to determine daily caloric and macronutrient targets [15].

The core challenge for these systems is to then recommend food items that both satisfy these nutritional constraints and appeal to the user's palate [13]. The typical pipeline involves a two-step process: first, an exclusionary filtering step removes all items that violate a user's "hard constraints," such as ingredients they are allergic to, specific dislikes, or dietary patterns like vegetarianism [15]. Second, the remaining pool of candidate items is ranked based on their alignment with the user's nutritional targets. This ranking is the crux of the recommendation logic, and its sophistication has evolved over time.

Early or simpler systems might focus only on a few key nutrients, such as calories or macronutrients (protein, fat, carbohydrates). However, more advanced research recognizes that a truly healthy diet is about more than just macros. These systems aim for a more holistic nutritional assessment by incorporating a much wider array of nutrients. For example, some systems evaluate the overall healthiness of a recipe by using a scoring system based on guidelines from public health bodies, such as the UK's Food Standards Agency (FSA), which assesses levels of sodium, fat, sugar, and saturated fat [13]. Our methodology aligns with this state-of-the-art approach by expanding the user and recipe nutritional vectors to include a comprehensive suite of essential micronutrients based on Recommended Dietary Allowances (RDA), thereby promoting meals that are not just calorically appropriate but holistically nutritious. To handle the increased complexity of these relationships, some recent work has explored the use of knowledge graphs and graph neural networks to model the intricate connections between users, recipes, ingredients, and nutrients, aiming to enhance the accuracy and context-awareness of health-based recommendations.

2.3. Knowledge Base Construction for Culturally-Specific FRS

The effectiveness of any FRS, particularly a health-aware one, is fundamentally contingent on the quality, comprehensiveness, and accuracy of its underlying knowledge base. This data foundation, which links food items to their nutritional properties, is a critical component that is often taken for granted in algorithmic-focused research. However, its construction presents significant data engineering challenges, especially when dealing with specific cultural cuisines.

A primary obstacle identified in the literature is the scarcity and poor quality of data for local and regional foods [20]. While comprehensive Food Composition Databases (FCDBs) exist for many Western countries, such resources for other parts of the world are often incomplete, outdated, or based on non-standardized analytical methods [22]. This lack of public, high-quality data forces many researchers to create their own

bespoke datasets, which complicates validation and hinders the reproducibility of their results [23].

Even when data is available, it often originates from heterogeneous sources—such as national food tables, international databases like the USDA's, and branded product labels—that are difficult to harmonize. Discrepancies in nutrient definitions, units of measurement, and data formats make merging these sources into a single, cohesive database a non-trivial task [20]. Our work directly confronts this harmonization challenge by developing a systematic pipeline to integrate the US-centric USDA database with the official Vietnamese FCT, standardizing ingredient names and nutrient columns to create a unified resource.

To populate the recipe side of the knowledge base, researchers are increasingly turning to unstructured sources like online blogs and digitized cookbooks. This necessitates the use of advanced AI and NLP techniques to transform raw text into structured, computable data [39]. This process often involves:

- **Text Extraction:** Using techniques like Optical Character Recognition (OCR) to convert scanned images of cookbooks into machine-readable text.
- **Information Extraction:** Employing NLP models, including modern LLMs, to parse the unstructured recipe text and extract key entities such as the recipe name, ingredient list, quantities, and preparation steps [40]. Our two-pass LLM-based approach for parsing recipes from PDF cookbooks is a state-of-the-art implementation of this principle.
- **Normalization and Standardization:** Applying linguistic and knowledge-driven processes to standardize the extracted data. This includes lemmatization (e.g., "eggs" to "egg") and synonym mapping (e.g., "plain flour" and "all-purpose flour" to a single canonical entity) to ensure consistent linkage between the recipe ingredients and the nutritional database. It also involves standardizing all ingredient quantities to a consistent unit of mass (grams) to enable accurate nutritional calculation.

The detailed articulation of this entire pipeline in our work is a significant contribution, as it provides a replicable methodology for overcoming the data barrier that has historically impeded FRS development for many of the world's cuisines.

2.4. Identifying the Actionability Gap

A critical analysis of the existing literature reveals that the dominant paradigm in health-aware FRS can be characterized as "passive matching." In this model, the system's role is to act as a sophisticated filter. It takes a user's profile and constraints, searches through a static database of recipes, and returns a ranked list of the "best-fitting" options [11]. The interaction typically ends there, leaving the user to select from this list. Our research, however, provides strong empirical evidence that this passive matching paradigm is fundamentally flawed and insufficient for the complex demands of personalized nutrition.

The experiments conducted during the development of our framework starkly illustrate this limitation. As we progressively increased the sophistication and realism of our nutritional matching model—from the simple L2 distance in V1, to the more nuanced weighted percentage deviation in V2, and finally to the holistic, RDA-inclusive model

in V3—the top match scores for pre-existing recipes consistently decreased [15]. The conclusion drawn from this trend is profound: it is highly improbable that a single, static recipe from a database will perfectly satisfy a user's complex and specific nutritional profile, which may include dozens of distinct macro- and micronutrient targets [15]. This finding exposes a critical "actionability gap": the disconnect between the best-available recommendation and the user's actual nutritional goal. Simply providing a list of imperfect matches is not enough to ensure a user can meet their health objectives.

The literature contains some nascent efforts to make recommendations more dynamic or actionable, but these fall short of closing the gap we have identified. Some systems propose recipe modifications, but this is typically limited to ingredient substitution, such as replacing an unhealthy ingredient with a healthier one or suggesting an alternative for an unavailable item [43]. This does not address the overall quantitative nutritional discrepancy of the final dish. Other systems are described as "dynamic" because they continuously update their recommendation models based on ongoing user feedback, such as new ratings or browsing history.[45] While this improves the quality of future recommendations, it does not provide the user with a solution for fixing the nutritional shortcomings of the current recommendation. The concept of providing "actionable advice" is recognized as a desirable research direction, but concrete, computationally-driven mechanisms for achieving it are rarely detailed in the literature.

Our work makes its most significant contribution by directly addressing this actionability gap. Our integrated methodology, with its novel "Nutrient Gap Supplementation" stage, transforms the FRS from a passive selector into an active dietary assistant. This represents a paradigm shift in the conceptual role of the system. Instead of merely acting as a selector that finds the best pre-existing entity from a database, our V4 system acts as a synthesizer. It constructs a new, composite meal solution that did not previously exist in the database, one that is explicitly designed to meet the user's complex needs. This is achieved through a unique, two-step computational process that occurs after the initial best-match recipe is identified:

1. **Recipe Portion Scaling:** The system first calculates an optimal scaling factor for the main recipe (e.g., 1.2x or 0.9x) to computationally minimize the overall nutritional deviation from the user's per-meal targets.
2. **Side Dish Supplementation:** If significant nutritional deficiencies remain even after optimal scaling, the system identifies the most critical nutrient deficit (e.g., protein) and queries its knowledge base for a complementary side dish that is dense in that specific nutrient. It then presents the user with a complete, two-part meal whose combined nutritional profile is a much closer match to their target.

This ability to not only identify a suitable base recipe but also to provide explicit, quantitative instructions on how to adapt and complete it is, to our knowledge, a novel approach in the field. It directly solves the problem that our own experiments uncovered: that a single recipe is rarely enough. By moving from passive matching to active, computational guidance, our framework represents a significant step toward creating FRS that are not just informative but truly practical and effective tools for personalized health management. Table 2 provides a comparative analysis of our final proposed

system against other state-of-the-art health-aware FRS, highlighting its unique contributions.

Table 2: Comparative Analysis of State-of-the-Art Health-Aware Food Recommendation Systems

System	Core Methodology	Health Constraints	Cultural Specificity	Recommendation Type (Key Differentiator)
Our Work	Content-Based + Nutrient Gap Analysis & Supplementation	User-defined Macro/Micro targets derived from BMR/TDEE and RDA.	**High** (Focused on Vietnamese Cuisine with a custom-built knowledge base).	**Active Guidance:** Provides a ranked recipe *plus* computational advice on portion scaling and side-dish supplementation to close nutritional gaps.
Yum-me [33]	Visual Quiz + Content-Based Filtering	User-defined nutritional expectations (e.g., calories, fat) and dietary restrictions (e.g., kosher, halal).	**Low** (General, with examples from American cuisine).	**Passive Match:** Provides a ranked list of recipes based on fine-grained visual preferences and high-level health filters.
FKGM [13]	Knowledge Graph + Multi-Task Learning	Healthiness score based on UK Food Standards Agency (FSA) ratings for sodium, fat, sugar, and saturated fat.	**Low** (General, uses Allrecipes dataset).	**Passive Match:** Provides a ranked list of recipes based on a learned healthiness score and user preference.
DIETOS [14]	Multi-Objective Optimization	Personalized healthiness score based on WHO guidelines, nutritional diversity.	**Low** (General).	**Passive Match:** Provides a ranked list of recipes that represent a Pareto-optimal solution across preference, health, and diversity objectives.

3 Methodologies

3.1 Knowledge Base Construction

The foundation of the recommendation system is a meticulously constructed knowledge base derived from three primary sources: Vietnamese cookbooks, the U.S. Department of Agriculture (USDA) Foundation Food dataset, and the 2007 Vietnamese Food Composition Table (FCT). The data processing pipeline was engineered to systematically extract, clean, standardize, and integrate information from these sources into a single, analysis-ready dataset.

The initial phase focused on extracting and structuring recipe information from several Vietnamese cookbooks, which were originally in PDF format. A PDF processing library was first used to acquire the raw text. Subsequently, a Large Language Model (LLM) was employed in a two-pass approach to manage the unstructured nature of the text. In the first pass, the LLM parsed the content to extract key semantic information, including the recipe name, serving size, cook time, and a list of ingredients with their original quantities. This extracted data then underwent a second LLM-driven standardization process. In this step, all ingredient quantities were uniformly converted to grams (g), ambiguous serving ranges were resolved to a single numerical value, and the total cook time was calculated in minutes by aggregating preparation and cooking times. The output of this stage was a clean, standardized recipe dataset with a consistent and computable format.

The second phase involved the construction of a comprehensive nutritional database. For the USDA Foundation Food dataset, which consists of multiple relational CSV files, a series of merge and filter operations were performed. These operations systematically consolidated the relational data, joining tables for food items, categories, and nutrient data to produce a single, coherent dataset. Critically, all nutrient values were standardized to a consistent baseline of a 100g serving, which is essential for accurate recipe calculations. The Vietnamese FCT required a more complex, multi-step process to extract data from its PDF format. After initial text extraction, regular expressions were used to isolate the nutritional tables. A custom script then parsed this content line-by-line, handling common formatting errors and character encoding issues. For ingredient names provided in both Vietnamese and English, only the English name was retained to ensure consistency with the USDA data.

The final stage of the pipeline was integration and harmonization. Here, the cleaned USDA and Vietnamese FCT datasets were merged into a unified nutritional database. This process involved de-duplicating entries and standardizing all nutrient names and units into a consistent schema, for instance, by mapping both "Lipid (g)" and "Total lipid (fat)" to a single "fat_g" column. Finally, the standardized recipe dataset was enriched by programmatically linking each ingredient to its corresponding entry in the unified nutritional database. This crucial matching step enabled the precise calculation of a full nutritional profile for every recipe, completing the construction of the system's knowledge base, which can be represented as a Recipe-Nutrient Matrix (R).

3.2. The Personalized Recommendation Engine

With the knowledge base established, the recommendation engine translates a user's needs into an optimized daily meal plan. This process is orchestrated by a central **Conversational AI Agent**, powered by a Large Language Model (LLM), which guides the user through two main stages: user-centric target modeling and a two-phase optimization algorithm. Agent receives input from user and based on information provided to generate suitable parameters for calling functions that calculate the user nutritional requirements.

3.2.1. User-Centric Nutritional Target Modeling

This stage converts a user's qualitative profile into a quantitative, multi-dimensional nutritional target vector (T_{user}). Instead of a static form, the agent engages in a dialogue to extract and standardize key user parameters. This allows for a more intuitive and flexible user experience. The critical information extracted includes:

- **Age**: The user's age in years.
- **Gender**: Male or Female.
- **Height**: The user's height in centimeters.
- **Weight**: The user's weight in kilograms.
- **Activity Level**: A categorical descriptor such as 'sedentary', 'moderately active', or 'very active'.
- **Health Objective**: The user's primary goal, such as 'weight loss', 'maintain weight', or 'muscle gain'.
- **Allergies**: A list of ingredients the user is allergic to (e.g., 'peanuts', 'shellfish').
- **Disliked Ingredients**: A list of ingredients the user dislikes but is not allergic to (e.g., 'coriander', 'ginger').
- **Max Prep Time**: The maximum time in minutes the user is willing to spend on a single meal, derived from phrases like "I need quick meals" or "I have about an hour for dinner".

Using these inputs, a hybrid approach is employed for maximum accuracy in target calculation. First, precise macronutrient and calorie targets are calculated using the Mifflin-St Jeor equation to determine the user's Basal Metabolic Rate (BMR).

$$BMR = (10 \cdot weight_{kg}) + (6.25 \cdot height_{cm}) - (5 \cdot \text{age}) + \text{s}$$

(where s is +5 for males and -161 for females)

This BMR is then adjusted by an activity factor to determine the Total Daily Energy Expenditure (TDEE), which represents the calories needed to maintain current weight. This value is further adjusted based on the user's health objective to set a final calorie goal.

$$TDEE = BMR \cdot ActivityFactor + GoalAdjustment$$

This final TDEE becomes the user's daily calorie target. It is then used to derive targets for protein, lipids, and glucids based on macronutrient ratios appropriate for the stated health objective.
Second, for the full spectrum of micronutrients (vitamins and minerals), the system references the Recommended Dietary Allowance (RDA) data. To ensure robust data linkage, a fuzzy string matching algorithm is used to dynamically map the nutrient names from the RDA source to the corresponding column names in the Recipe-Nutrient Matrix. This automated approach avoids errors from manual mapping and makes the system resilient to minor variations in data schemas. The final output is a comprehensive target vector, T_{user}, that defines the ideal daily intake for all available nutrients.

3.2.2. Two-Phase Meal Plan Optimization

Based on the nutricient requirements that created from the module above, the recommendation will provide suitable food candidates that meet the requirement. The core of the recommendation engine is a two-phase optimization process designed to navigate the high-dimensional search space of meal combinations efficiently. This approach is necessary to balance the computational complexity of meal planning with the need for holistic nutritional accuracy.

3.2.2.1. Macro-Focused Dynamic Programming

The first phase addresses the combinatorial explosion inherent in meal planning by using Dynamic Programming (DP) to find a large set of candidate meal plans that are well-optimized for the four primary macronutrients. This focus is crucial for managing complexity, as simultaneously optimizing for all ~100 nutrients is computationally infeasible—a problem known as the "curse of dimensionality".
The process begins by discretizing the continuous nutrient values into a finite number of bins to create a manageable state space. The DP algorithm then iteratively builds 3-meal plans ($k = 1,2,3$), minimizing a cumulative deviation score. The cost of adding a recipe (r) to a specific meal (k) is calculated as its weighted deviation from that meal's designated nutritional targets (e.g., breakfast might be allocated 25% of the total daily calories). This cost function, $d_k(r)$, is defined as:

$$d_k(r) = \sum_{j \in Macros} w_j \cdot \frac{|R_{rj} - T_{j,meal_k}|}{T_{j,meal_k}} \quad (1)$$

where S_k is the state vector of nutrient bins for meal k. The DP algorithm uses a recurrence relation to find the optimal (lowest cost) way to reach a given nutritional state after k meals, building upon the optimal solutions found for $k - 1$ meals. The relation is expressed as:

$$DP(k, S_k) = min_{r \in Recipes} \left[DP(k - 1, S_k - S_r) + d_k(r)\right] \quad (2)$$

where S_k is the state vector of nutrient bins for meal k. This phase effectively narrows the search space from millions of potential combinations to tens of thousands of viable candidates that satisfy the user's main energy and macro requirements.

3.2.2.2. Holistic Scoring and Final Selection

The second phase evaluates every candidate plan generated by the DP against the user's full nutritional target vector T_{user}. A holistic scoring function calculates a final "match score" for each plan. This score is a weighted sum that primarily reflects the plan's deviation from macronutrient targets but also incorporates a significant penalty for any shortfalls in micronutrient coverage. The final score for a plan is calculated as:

$$Score_{final} = Score_{macro} + (w_{micro} \cdot Penalty_{micro}), \quad (3)$$

The macro score component measures the accuracy against the user's primary goals for the entire day:

$$Score_{macro} = \sum_{j \in Macros} w_j \cdot \frac{|\sum_{r \in Plan} R_{rj} - T_j|}{T_j} \quad (4)$$

The micro penalty component aggregates the proportional deficit for each micronutrient. This ensures that the system prioritizes nutritionally complete plans over those that may be perfect on macros but deficient in essential vitamins and minerals.

$$Pelnaty_{micro} = \frac{1}{M} \sum_{m \in Micros} max(0, \frac{T_m - \sum_{r \in Plan} R_{rm}}{T_m}) \quad (5)$$

The plan with the lowest final match score is selected as the optimal recommendation. This two-phase strategy balances computational efficiency with nutritional comprehensiveness, yielding a highly personalized and well-rounded meal plan.

3.2.3. Conversational Feedback and Iteration

Following the optimization, the process returns to the Conversational AI Agent, which now acts as a communicative and adaptive presenter.

Result Synthesis and Presentation: The agent takes the final meal plan (a list of recipe IDs). It queries the Recipe Database to retrieve full recipe details, including ingredients, quantities, and cooking instructions. It then presents this complete, human-readable plan to the user, explaining why this plan was chosen (e.g., "This plan meets your protein goals while staying within your calorie target for weight loss."). It stands by to answer any follow-up questions the user might have about substitutions or preparation.

Feedback Loop and Continual Learning: The agent's final task is to gather feedback on the proposed meal plan. It prompts the user for a rating (e.g., "How do you feel about this plan? Great/Okay/Not for me"). It also asks the user to confirm which meals they consumed. This feedback is crucial:

- User ratings (positive, neutral, or negative) are logged and associated with the suggested recipes.
- The record of consumed meals is used to update the user's food history.
- This historical data and preference information will be used as a new input in future recommendation cycles, allowing the system to learn from past interactions and provide increasingly personalized and accurate suggestions over time. If no feedback is provided, a "neutral" rating is assumed.

4 Results and Discussions

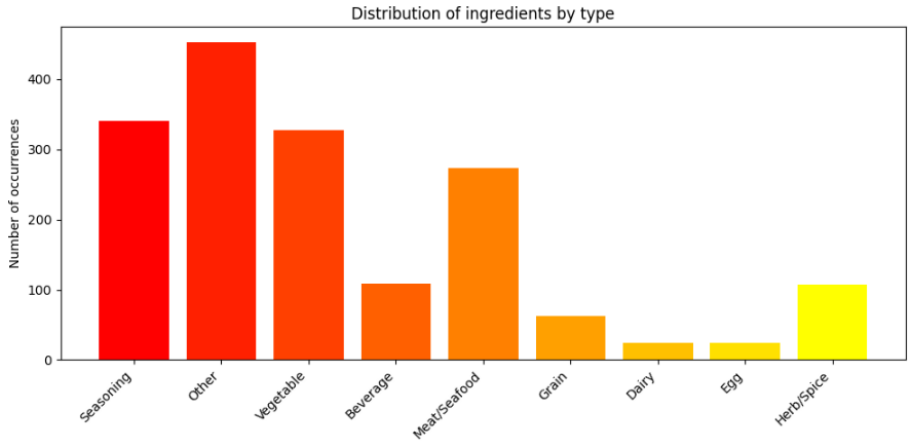

Figure 1: Distribution of ingredients, by type

The data processing pipeline culminated in a rich dataset of 143 recipes featuring 210 unique ingredients, which were used in various combinations. An analysis of ingredient frequency revealed patterns highly characteristic of Vietnamese cuisine. Vegetables were the most frequently used category, appearing 585 times, followed by the 'Other' category (including sauces and seasonings) at 368 times and Meat at 353 times. Grains and Legumes appeared 77 times, while Dairy products were the least frequent at just 14 times. The most popular ingredients within the main food groups were Chicken, Shrimp, and Beef for Meat; Coriander, Shallots, and Lemon for Vegetables; and Rice, Vermicelli, and Flour for Grains/Legumes.

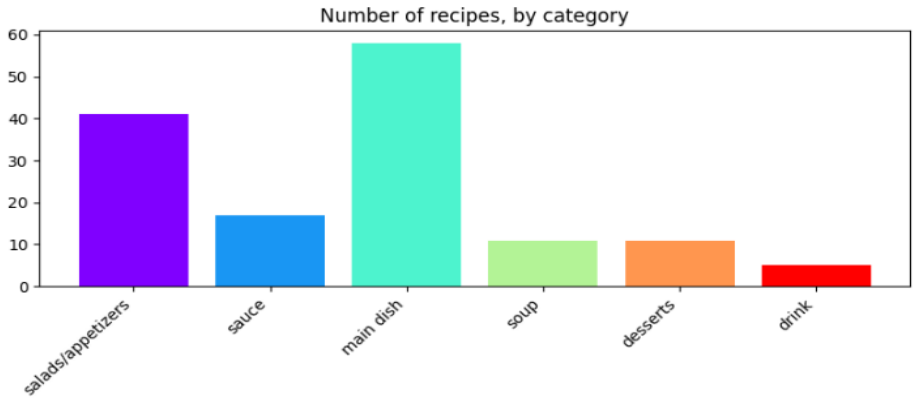

Figure 2: Number of recipes, by category

Among the 143 recipes, main dishes, comprised of rice, meat dishes, noodles, etc... made up the largest group. The second most frequent category was salads/appetizers. Sauce appeared around 20 times, underscoring the diversity of spices used in

Vietnamese cuisine. Soups, desserts, and drinks were less frequent, but contributed to the dataset's overall variety.

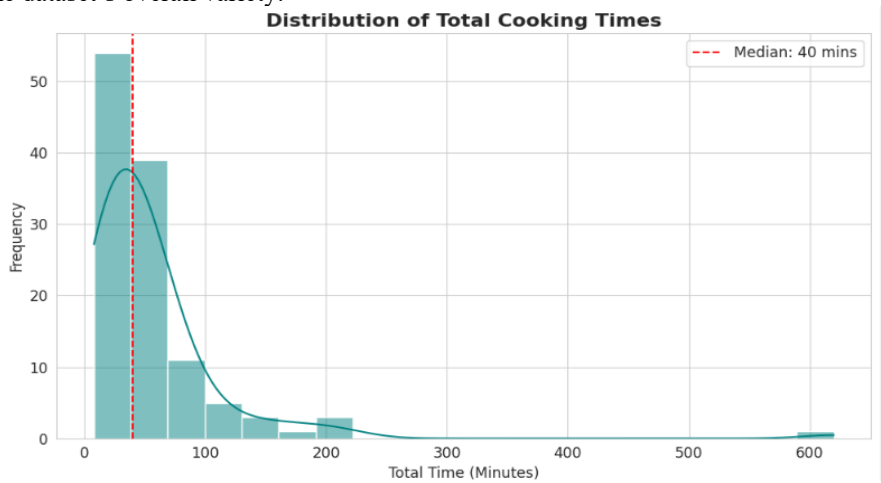

Figure 3: Distribution of Total cooking times

Most recipes are medium-easy to make, with their cooking time varying around 40 minutes, making them more accessible to home cooks. This reflects the suitability of the dishes in the datasets for home-ready meals, being flavorful and practical, not requiring extensive preparation time.

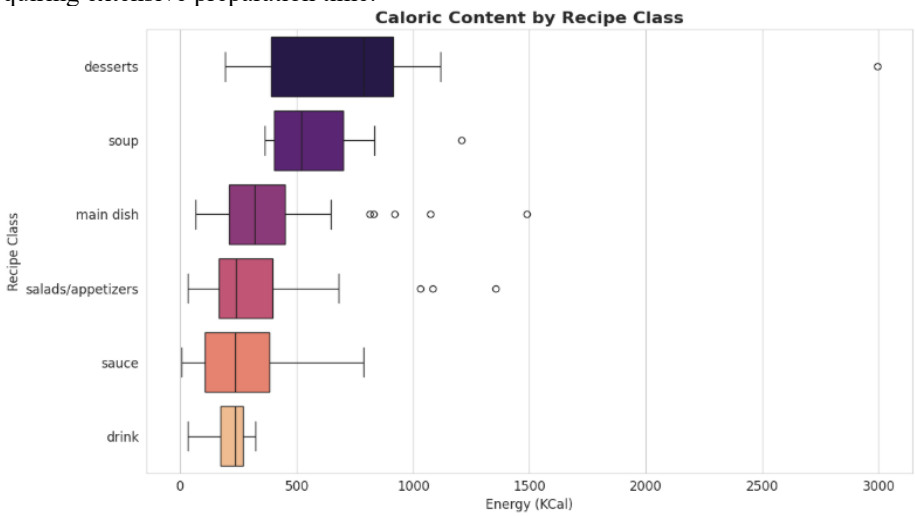

Figure 4: Caloric content, by recipe class

Based on calorie content, desserts usually have the most energy in one dish. Next is soup, main dish, salads/appetizers, then sauces and drinks. Dishes show a wide variety of calorie values and numerous outliers, showing that the dishes range from light meals to heavy ones. One should balance the caloric content of different types of dishes to ensure enough energy.

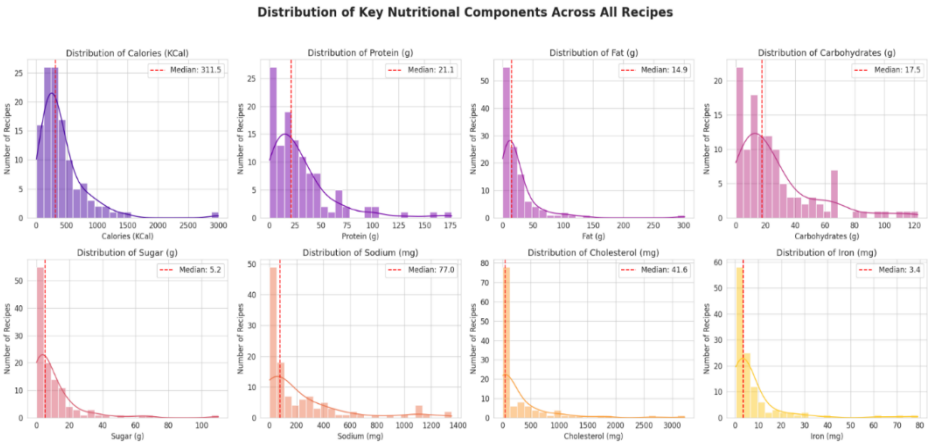

Figure 5: Distribution of Key nutritional component across all recipes

The distribution of energy is heavily concentrated below 750 KCal, with a median of 311.5 KCal. A typical recipe is quite moderate in calories. All recipes' nutrients are skewed, showing the imbalance in the ingredients' macronutrients. Concerning micro-nutrients, low sugar content recipes is extremely concentrated, with a median of only 5.2 grams. Like macronutrients, micronutrients are also skewed, but for Iron, the skewness is less than sodium or cholesterol. This shows that foods are imbalanced in one or more key nutrients. Our system should consider nutrient compensation, helping user balance their nutrients intake across meals, rather than relying on one meal a day to made up the whole required nutritional content.

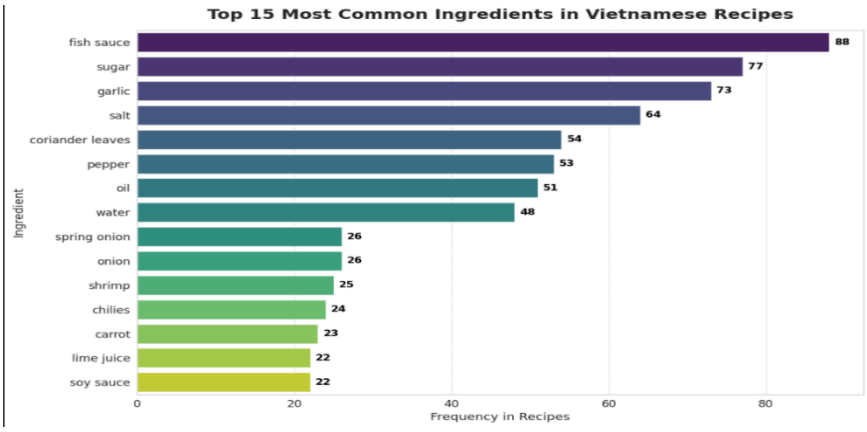

Figure 6: 15 Most common ingredients

Fish sauce, sugar and garlic are the 3 most used ingredients for cooking. Salt and vegetable oil, even though being used in most dishes, only have 64 and 51 uses. Shrimp

was the only type of meat was in top 15, and most ingredients are vegetables, spice, fruits and oils.

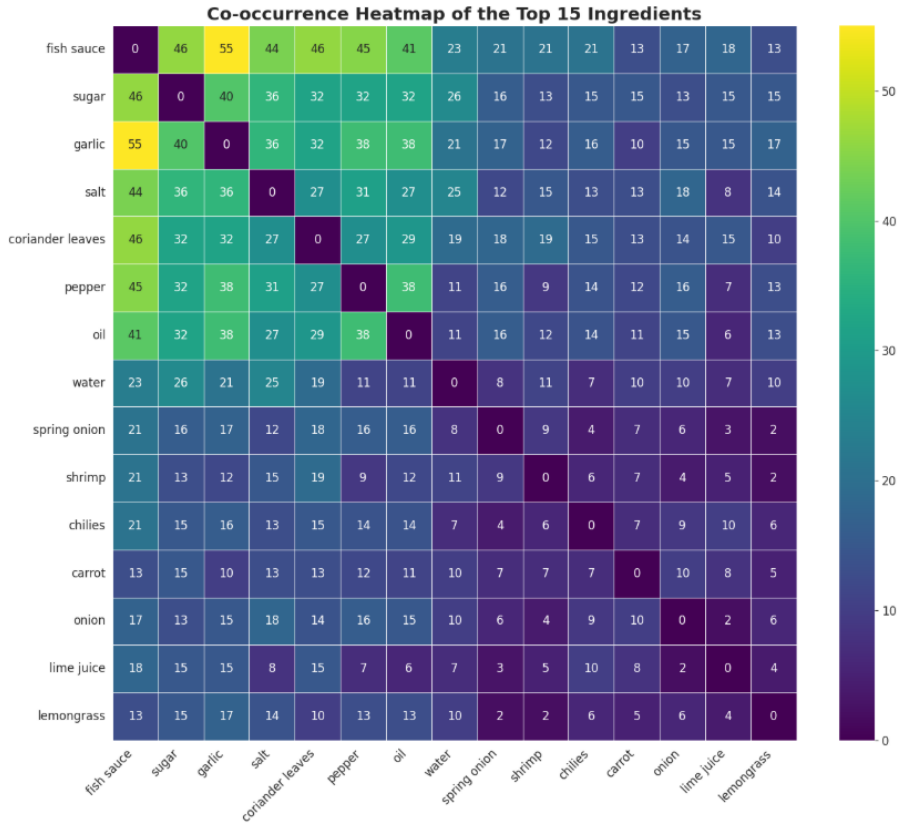

Figure 7: Co-occurence heatmap of top 15 ingredients

Considering the association rule amongst the ingredients, fish sauce and sugar goes with each other the most, with 55 times on 157 recipes. Spices usually go with each other, and especially savory-sweet-aromatic, with fish sauce and sugar also dominate the heatmap with 46 occurrences.

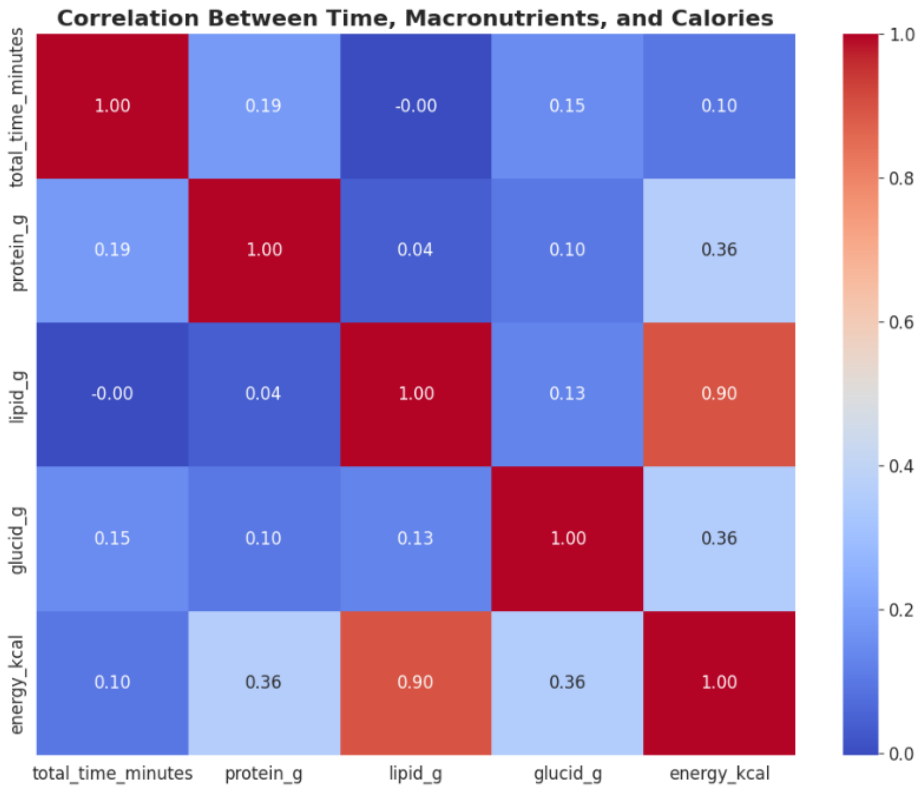

Figure 8: Correlation between time, macronutrients, and calories

Fat is the primary deliver of calories with a correlation of +0.90, while cooking time is a poor predictor. Food that cooks for long usually have low calories. Protein and carbohydrates show an identical positive correlation with calories. Macronutrients have no strong correlation between each other. Users should eat a wide variety of food to ensure sufficient nutrients, rather than based on the core nutrients to base other nutrients.

An experiment designed to test the first three methodologies involved 20 users with diverse profiles interacting with the 143 recipes. The results, measured by a **Match Score**, showed a clear trend. The top single match_score for Method 2 was 82.12, which dropped to 64.2 for Method 3, and further to 3.87 for Method 1. This shows that by updating metric, the accuracy has been more appropriate but still has a struggle when increasing the number of nutrition types that need to be recommended. This led to the important conclusion that a simple similarity metric is insufficient for this task and that, crucially, no single food is likely to satisfy all of a person's nutritional requirements without adjustment. This finding strongly validates the necessity of a more advanced approach. Consequently, Methodology Version 4 was developed, shifting the paradigm from recommending single, imperfectly matched recipes to constructing holistic, optimized daily meal plans using Dynamic Programming. This advanced methodology was tested against the same cohort of 20 users.

The adoption of Dynamic Programming marked a significant increase in the complexity of the model. This was reflected in the **Processing Time**, which became a key metric for evaluating computational feasibility. The system successfully generated a 3-meal plan for all 20 users, but the average time required to compute an optimal plan rose to **64.63 seconds** per user. This substantial increase is a direct consequence of the algorithm's exhaustive search through a vast state space, defined by the number of meals, recipes, and nutritional bins, to guarantee an optimized solution.

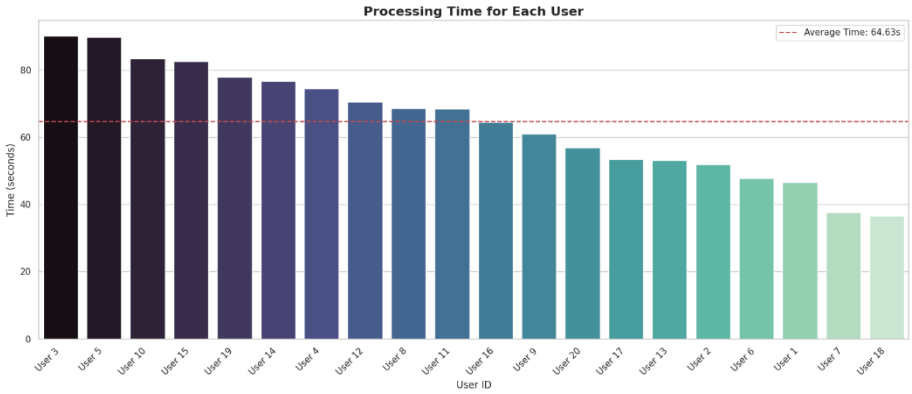

Figure 9: The processing time of each user

The processing time exhibits considerable variance, ranging from approximately 37 seconds to approximately 90 seconds (about 3 minutes). This variation is primarily influenced by the number of candidate recipes available after the initial filtering stage. User profiles with fewer allergies or dietary restrictions resulted in a larger pool of recipes, thereby expanding the search space for the algorithm and increasing computation time. While the duration is notable, it demonstrates the computational trade-off required to achieve a high degree of personalization and nutritional accuracy, a task that simpler, faster methods failed to accomplish.

Despite the computational cost, the quality of the recommendations saw a dramatic improvement. As illustrated in **Figure 10: Match Score for Each User**, the system consistently delivered high-quality meal plans. The average Match Score across all users was **86.03**. The performance was remarkably consistent, with the lowest score being a robust 77.4. This narrow range of high scores indicates that the DP algorithm is highly reliable and capable of finding effective meal combinations for nearly every profile, a significant leap from the inconsistent performance of previous methodologies.

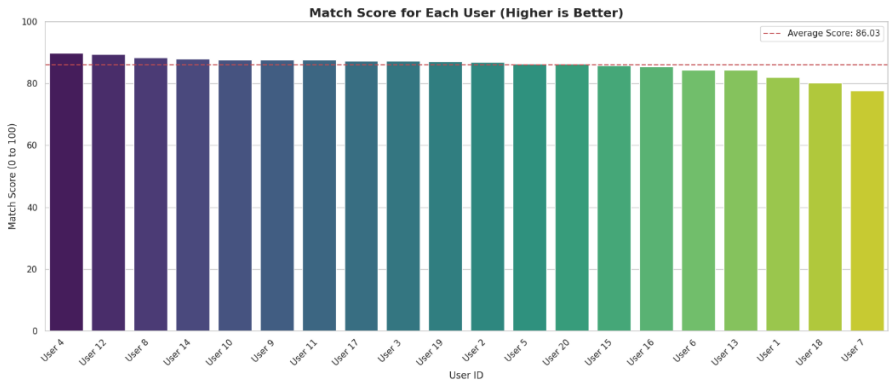

Figure 10: The match score for each user

An analysis of performance across different health objectives, as shown in **Figure 11: Match Score Comparison by Health Objective**, reveals the model's adaptability. Users with 'Weight Loss' and 'Maintain Weight' objectives generally achieved the highest and most consistent scores. This is likely due to the abundance of moderate-calorie recipes in the knowledge base, which provides the algorithm with greater flexibility. The 'Muscle Gain' objective presented a greater challenge, requiring high protein intake which is less common in single dishes. This resulted in a slightly lower median score and wider score variance for this group, highlighting the difficulty of satisfying this specific nutritional profile. Nevertheless, the system still produced highly relevant plans, demonstrating its effectiveness even under tight constraints.

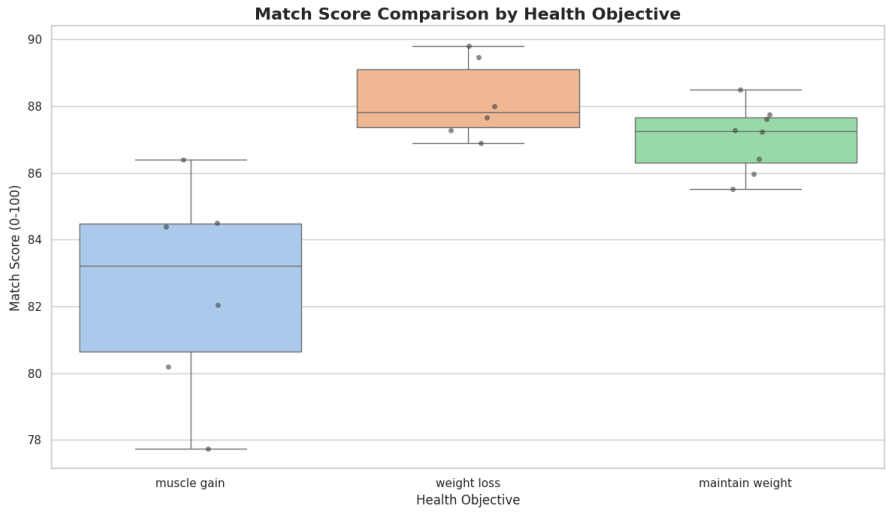

Figure ***11:*** *The match score based on individual index*

The results from Methodology 4 confirm that a sophisticated, optimization-based approach is essential for the complex task of personalized meal planning. While the introduction of Dynamic Programming significantly increased the average processing time to **64.63 seconds**, this computational investment yielded a profound improvement in recommendation quality and reliability, with an average Match Score of **85.2**.

The system demonstrated its robustness by successfully generating tailored plans for 100% of the diverse user profiles. The analysis revealed that the algorithm's performance is sensitive to the constraints of the user's health objective and the size of the available recipe pool, which directly impacts computational load. The qualitative case study further validates the model's primary advantage: its ability to construct a holistically balanced daily diet by synergistically combining different meals, a feat unattainable by single-recipe recommendation methods.

In conclusion, this research establishes a strong case for using optimization algorithms in nutritional recommendation systems. The trade-off between computational time and recommendation accuracy is a critical consideration, and Methodology 4 proves that investing in computational complexity can lead to vastly superior and more practical outcomes for the end-user. Future work should explore heuristic-based optimization techniques (e.g., Genetic Algorithms) that may offer a more balanced compromise between speed and optimality, potentially reducing processing times while maintaining a high level of personalization.

5 Conclusion

This paper successfully addressed the critical challenge of creating a culturally-aware and scientifically-grounded Food Recommendation System (FRS) capable of generating complete, personalized daily meal plans. We demonstrated a comprehensive framework that moves beyond conventional single-recipe suggestions to tackle the more complex problem of holistic dietary planning, with a specific focus on Vietnamese cuisine.

Our primary contributions are threefold. First, we established a robust and replicable data engineering pipeline that leverages Large Language Models to construct a high-quality knowledge base from unstructured and heterogeneous sources, a foundational step that makes an under-resourced cuisine computationally accessible. Second, we grounded our system in established nutritional science by implementing a user-profiling module that calculates precise dietary targets using the Mifflin-St Jeor equation for BMR/TDEE and integrating RDA standards for comprehensive micronutrient coverage. Finally, we introduced a novel two-phase optimization algorithm, using Dynamic Programming, that efficiently navigates the immense combinatorial search space to synthesize 3-meal daily plans. This core technical innovation effectively solves the "planning gap" by constructing meal plans that are not only calorically appropriate but also holistically balanced in both macro- and micronutrients.

The results confirm the efficacy of our approach. While computationally more intensive than simpler methods, our framework consistently produces high-quality, reliable meal plans for diverse user profiles, achieving an average match score of 86.03. This demonstrates a significant improvement in recommendation quality and validates

the necessity of sophisticated optimization techniques for the complex demands of personalized nutrition.

3not just informative, but are truly practical and effective tools for personal health management.

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